

A Synthetic Multi-Modal Variable to Capture Cardiovascular Responses to Acute Mental Stress and Transcutaneous Median Nerve Stimulation

Yuan Yuan Zhou, Jesse D. Parreira, Sina Masoumi Shahrababak, Jesus Antonio Sanchez-Perez, Farhan N. Rahman, Asim H. Gazi, Omer T. Inan, *Fellow, IEEE* and Jin-Oh Hahn*, *Senior Member, IEEE*

Abstract— Objective: To develop a novel synthetic multi-modal variable capable of capturing cardiovascular responses to acute mental stress and the stress-mitigating effect of transcutaneous median nerve stimulation (TMNS), as an initial step toward the overarching goal of enabling closed-loop controlled mitigation of the physiological response to acute mental stress. **Methods:** Using data collected from 40 experiments in 20 participants involving acute mental stress and TMNS, we examined the ability of six plausibly explainable physio-markers to capture cardiovascular responses to acute mental stress and TMNS. Then, we developed a novel synthetic multi-modal variable by fusing the six physio-markers based on numerical optimization and compared its ability to capture cardiovascular responses to acute mental stress and TMNS against the six physio-markers in isolation. **Results:** The synthetic multi-modal variable showed explainable responses to acute mental stress and TMNS in more experiments (24 vs ≤ 19). It also exhibited superior consistency, balanced sensitivity, and robustness compared to individual physio-markers. **Conclusion:** The results showed the promise of the synthetic multi-modal variable as a means to measure cardiovascular responses to acute mental stress and TMNS. However, the results also suggested the potential necessity to develop a personalized synthetic multi-modal variable. **Significance:** The findings of this work may inform the realization of TMNS-enabled closed-loop control systems for the mitigation of sympathetic arousal to acute mental stress by leveraging physiological measurements that can readily be implemented in wearable form factors.

Index Terms— Acute mental stress, transcutaneous median nerve stimulation, peripheral neuromodulation, cardiovascular physio-marker, closed-loop control

I. INTRODUCTION

ACUTE mental stress arises from various sources including work-related pressure, financial crises, social relationships, and, more recently, isolation [1], [2], [3], [4], [5], [6]. Acute mental stress profoundly deteriorates the quality of life across multiple dimensions, encompassing physical, emotional, cognitive, and social to list a few [1], [4], [5], [6], [7], [8], [9]. Frequent exposure to acute mental stress can result in an increased risk of physical and mental health disorders, e.g., cardiovascular disease, depression, and post-traumatic stress disorder [8], [9], [10], [11], [12], [13]. Hence, effective management of acute mental stress is a high societal priority.

Acute mental stress elicits cardiovascular responses. Prior work (including our own) has shown that acute mental stress elicits sympathetic arousal including (i) an increase in heart rate (HR) [1], [14], [15], [16]; (ii) a decrease in pulse transit time (PTT) [14], [16]; (iii) a decrease in pulse arrival time (PAT) [14], [17]; (iv) a decrease in pre-ejection period (PEP) [14], [18]; (v) an increase in left ventricular ejection time (LVET) [14]; and (vi) an increase in systolic and diastolic blood pressure (BP) [10], [14], [19]. These changes may occur in many subjects but not in every individual subject. Still, there is an opportunity to monitor and mitigate sympathetic arousal to acute mental stress by exploiting these cardiovascular physio-markers.

Transcutaneous peripheral neuromodulation has garnered attention as a promising approach to non-invasively influence neural activities and physiological processes beyond conventional pharmacological interventions [20], [21], [22], [23], [24], [25], [26], [27], [28]. Notably, there is an increasing body of evidence that transcutaneous peripheral neuromodulation (including stimulation of the median nerve, auricular branch of the vagus nerve, and cervical vagus nerve) may help mitigate the sympathetic arousal due to acute mental stress [20], [21], [22], [24]. In particular, transcutaneous median nerve stimulation (TMNS) stands out as a promising candidate for non-invasive and ubiquitous mitigation of sympathetic arousal to acute mental stress by virtue of its unique advantages: (i) median nerves are richly connected to the central nervous system and the stress-related pathways, thereby enabling active modulation of autonomic function to mitigate sympathetic arousal to acute mental stress and to

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Y. Zhou, J.D. Parreira, S. Masoumi Shahrababak, and J.O. Hahn are with the Department of Mechanical Engineering, University of Maryland, College Park, MD 20742, USA (correspondence e-mail: jhahn12@umd.edu).

J.A. Sanchez-Perez, F.H. Rahman, and O.T. Inan are with the School of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA 30332, USA.

A.H. Gazi is with the Computer Science Program, Harvard University, Allston, MA 02134, USA.

promote calmness and relaxation [22], [25]; and (ii) median nerves can be stimulated easily at the wrist, enabling minimally obtrusive, ultra-convenient intervention using wrist wearables (e.g., a wristwatch) [22], [29].

Prior work has demonstrated that TMNS may lower BP [29], [30] and HR [31]. Hence, it may be possible to capture the effect of TMNS via physio-markers representing various cardiovascular responses. However, the best method to capture the stress-mitigating effect of TMNS is unknown. This knowledge gap remains a major hurdle to overcome toward the realization of a TMNS-enabled closed-loop control paradigm for autonomous and active mitigation of sympathetic arousal to acute mental stress. One reasonable hypothesis is that the stress-mitigating effect of TMNS may be captured by the physio-markers which respond sensitively to acute mental stress. This hypothesis appears plausible, as one would expect that if acute mental stress responses are blunted by TMNS, then the sympathetic arousal in the physio-markers induced by acute mental stress will also be normalized.

As an initial step toward the overarching goal of enabling closed-loop controlled mitigation of sympathetic arousal to acute mental stress, we developed a synthetic multi-modal variable capable of capturing cardiovascular responses to acute mental stress and the stress-mitigating effect of TMNS. Specifically, we focused on utilizing physiological signals compatible with a potential wearable-based implementation. For this purpose, we considered the six plausibly explainable physio-markers responsive to acute mental stress: HR ($\uparrow$), PTT ($\downarrow$) and PAT ($\downarrow$) for diastolic and systolic BP ($\uparrow$), PEP ($\downarrow$) for stroke volume (SV) ($\uparrow$), photoplethysmogram (PPG) amplitude ($\downarrow$) for total peripheral resistance (TPR) ($\uparrow$), and the product of HR and LVET ($\uparrow$) for cardiac output (CO) ($\uparrow$), where the arrows indicate the expected directions of change. Using data collected from 40 experiments in 20 participants involving acute mental stress and TMNS, we examined the ability of these physio-markers to capture cardiovascular responses to acute mental stress and TMNS. Then, we developed a novel synthetic multi-modal variable by fusing these six physio-markers based on numerical optimization and compared its ability to capture cardiovascular responses to acute mental stress and TMNS against the individual physio-markers in isolation. In addition, in an effort to probe the potential need for a personalized synthetic multi-modal variable, we compared population-average vs personalized synthetic multi-modal variables developed by the proposed approach.

This paper is organized as follows. Section II describes the experimental data, its pre-processing details, and our approach to develop a synthetic multi-modal variable from six plausibly explainable physio-markers. Section III describes key results, which are discussed in Section IV. Section V concludes the paper with potential future directions.

II. METHODS

A. Experimental Data and Pre-Processing

We used experimental data collected from 20 young healthy

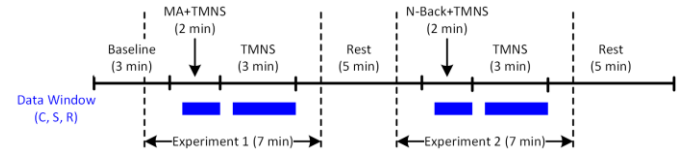


Fig. 1. Experimental protocol and data window used to calculate consistency, balanced sensitivity, and robustness metrics. MA: mental arithmetic. TMNS: transcutaneous median nerve stimulation. C: consistency. S: balanced sensitivity. R: robustness.

participants in our prior work under the approval of the Institutional Review Board at Georgia Institute of Technology (H18452) and the Navy Human Research Protection Office [22]. Each subject underwent a study protocol shown in Fig. 1, which consisted of (i) a baseline period (3 min), (ii) mental arithmetic and TMNS (2 min), (iii) TMNS alone (3 min), (iv) a rest period (5 min), (v) N-back and TMNS (2 min), (vi) TMNS alone (3 min), and (vii) a rest period (5 min). Throughout the study protocol, the following physiological signals were recorded continuously at a 2 kHz sampling rate: (i) electrocardiogram (ECG; BN-RSPEC, Biopac Systems, Goleta, CA, USA), (ii) transmissive PPG sensed at a fingertip (BST09001S, Shanghai Berry Electronic Tech, Shanghai, China interfaced with PPG100C, Biopac Systems, Goleta, CA, USA), and (iii) seismocardiogram (SCG) sensed at the chest using a custom accelerator module. These three physiological signals have been described extensively in the literature, but briefly, they can be ascribed to the following physiological origins. The ECG represents the electrical rhythm of the heart and is measured with electrodes placed on the surface of the chest. The PPG captures the blood volume pulse signal and is measured using a light source (e.g., LED) and a sensor (e.g., photodiode) placed on the skin, such as on a digit, earlobe, or the wrist. The SCG is a measure of the chest wall vibrations in response to the movement of the heart and the pumping of blood through the vasculature. These physiological signals were leveraged in this work because of their ability to capture a combination of electrical and mechanical aspects of cardiac function (i.e., through pairing the ECG with the SCG) and both cardiac and vascular characteristics (i.e., through pairing the ECG and the SCG with the PPG). TMNS was delivered using the DS8R Current Stimulator (Digitimer, Broadway, Letchworth Garden City, UK).

The data analysis procedure used in this work is shown in Fig. 2. We pre-processed the physiological signals as described in our prior work [14], [22]. In brief, we filtered the physiological signal waveforms to remove low-frequency drift and high-frequency noise. Then, we localized and extracted the R waves in the ECG. Then, we gated individual PPG and SCG beats using these R waves. Subsequently, we extracted fiducial points in the physiological signals on a cardiac beat-by-beat basis: (i) the timings and values of the maximum and minimum peaks as well as the foot in the PPG, and (ii) the timings of the AO point (a signature of aortic valve opening) and the AC point (a signature of aortic valve closure) in the SCG.

Then, we extracted the six plausibly explainable physio-markers on a cardiac beat-by-beat basis as follows. First, we

extracted instantaneous HR as the reciprocal of heart period, defined as the time interval between the neighboring R waves in the ECG. Second, we extracted PTT as the time interval between the AO point in the SCG and the foot (determined by the intersecting tangent method [32]) in the PPG. Third, we extracted PAT as the time interval between the R wave in the ECG and the foot in the PPG. Fourth, we extracted PEP as the time interval between the R wave in the ECG and the AO point in the SCG [33]. Fifth, we extracted PPG amplitude as the peak-to-peak amplitude of the PPG beat. Sixth, we extracted LVET as the time interval between the AO point and

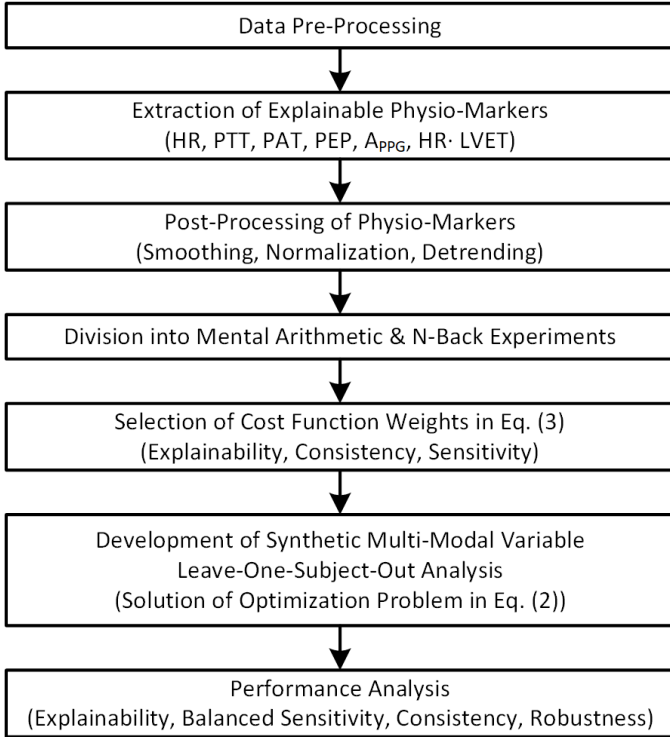


Fig. 2. Data analysis procedure. The problem of developing a synthetic multi-modal variable was cast into a constrained optimization problem. The tuple of weights in the cost function used in the optimization problem was determined to result in the synthetic multi-modal variable which (i) showed explainable responses to acute mental stress and transcutaneous median nerve stimulation (TMNS) in many experiments and (ii) harmoniously maximized the following metrics: consistency and sensitivity pertaining to combined acute mental stress and TMNS responses as well as TMNS response alone. Then, leave-one-subject-out analysis was conducted using the optimal tuple of cost function weights to develop and evaluate the ability of the synthetic multi-modal variable to capture cardiovascular responses to acute mental stress and TMNS. A_{PPG} : PPG amplitude.

the AC point in the SCG [33]. Note that these physio-markers are plausibly explainable: (i) PTT and PAT are inversely proportional to diastolic and systolic BP in acute settings, respectively [32]; (ii) PEP is inversely proportional to cardiac contractility and thus to SV [34], [35]; (iii) PPG amplitude at the extremity locations is inversely proportional to the degree of vasoconstriction (namely, TPR) [36], [37]; and (iv) LVET is proportional to SV [34], [38].

We resampled the physio-markers at 1Hz sampling rate. Then, we post-processed the physio-markers on a participant-by-participant basis as follows so as to retain the legitimate physiological variability therein while suppressing the

erroneous beat-by-beat variability incurred by, e.g., the errors introduced by the inaccuracies associated with the detection of fiducial points in the physiological signals required to extract the physio-markers. First, we smoothed the physio-markers via 30s moving average filtering. Second, we normalized each physio-marker via z-score normalization using its mean and standard deviation across the entire experimental protocol (Fig. 1). Third, we detrended the physio-markers to remove slow drifts not relevant to acute responses, if any, as follows: (i) we removed the average values pertaining to the baseline period and the two rest periods from the respective baseline and rest periods; and (ii) we removed the linear interpolation of the baseline and/or the rest periods before and after each acute mental stress-TMNS and TMNS period (shown as “Experiment” in Fig. 1) from the corresponding period. Fourth, we extracted two experiments from the resulting time series sequences of the physio-markers, each of which consists of an acute mental stress challenge (mental arithmetic or N-back challenge) and TMNS, TMNS alone, and 1-min periods before and after them as depicted in Fig. 1. We used the resulting 40 experiments to conduct the subsequent analysis outlined in Sections II.B-II.D (Fig. 2).

B. Development of Synthetic Multi-Modal Variable

Our approach to capturing cardiovascular responses to acute mental stress and the stress-mitigating effect of TMNS is to optimally combine the six plausibly explainable physio-markers (namely, HR, PTT, PAT, PEP, PPG amplitude, and $HR \cdot LVET$) into a novel synthetic multi-modal variable. We in particular sought to develop a synthetic multi-modal variable in the form of a linear combination of the above-mentioned plausibly explainable physio-markers (which is a simple yet effective way to probe the initial proof-of-concept of our approach before more sophisticated regression methods are conceived):

$$\phi(y(t)) = \sum_{i=1}^{N_y} \alpha_i y_i(t) \quad (1)$$

where $\phi(y(t))$ is the synthetic multi-modal variable at a time instant t , $y(t) = \{y_i(t) | i = 1, \dots, N_y\}$ is the vector of physio-markers at a time instant t , $N_y = 6$ is the number of physio-markers to be included, and α_i , $i = 1, \dots, N_y$ are the weights pertaining to individual physio-markers. To standardize the direction of changes in the physio-markers to acute mental stress and TMNS, we defined our physio-markers as follows: $y = \{HR, PTT^{-1}, PAT^{-1}, PEP^{-1}, A_{PPG}^{-1}, HR \cdot LVET\}$ where A_{PPG} is PPG amplitude. In this way, all the physio-markers are expected to increase in response to acute mental stress and decrease in response to TMNS, although in reality the individual physio-marker responses may exhibit a large inter-individual variability and may even show unexplainable responses (e.g., they may decrease in response to acute mental stress and/or increase in response to TMNS). Then, the problem of developing a synthetic multi-modal variable reduces to the problem of finding optimal weights assigned to the physio-markers when combining them. We formulated a standard constrained optimization problem below to solve this problem:

$$\min_{\alpha_i, 1 \leq i \leq N_y} J, \text{ s. t. } \sum_{i=1}^{N_y} \alpha_i^2 = 1, 0 \leq \alpha_i \leq \Delta \quad (2)$$

where the equality constraint intends to standardize the magnitude of the weight vector (by specifying its 2-norm to unity) while the inequality constraints intend to foster the contribution of all the physio-markers (by limiting the magnitude of individual weights to within Δ , set to 0.9 in this work). By enforcing the weights to be positive, the synthetic multi-modal variable is expected to ideally exhibit a positive response to acute mental stress and a negative response to TMNS. We defined the cost function J as follows:

$$J = w_1 J_1 + w_2 J_2 + w_3 J_3 \quad (3)$$

where J_1 , J_2 , and J_3 are the cost functions pertaining to baseline/rest period (J_1), acute mental stress-TMNS period (J_2), and TMNS period (J_3) in the experimental protocol, respectively (see Fig. 1), and w_1 , w_2 , and w_3 are the corresponding weights. We defined these cost functions as follows. In the baseline/rest period, the synthetic multi-modal variable should ideally show zero response. Hence, we defined J_1 as follows:

$$J_1 = \frac{1}{N_1} \sum_{k=1}^{N_1} \phi^2(y(k)) \quad (4)$$

where N_1 is the number of data points in the baseline/rest period. In the presence of acute mental stress and TMNS, our experimental results (to be shown in Section III) showed that cardiovascular responses to acute mental stress are generally larger in magnitude than those to TMNS. Hence, in the period when both acute mental stress and TMNS are present, the synthetic multi-modal variable should ideally show positive response, since all the physio-markers are expected to increase and all the physio-marker weights α_i , $i = 1, \dots, N_y$ are greater than or equal to zero. Hence, we defined J_2 as follows:

$$J_2 = \frac{1}{N_2} \sum_{k=1}^{N_2} -\text{sgn}(\phi(y(k))) \phi^2(y(k)) \quad (5)$$

where N_2 is the number of data points in the acute mental stress-TMNS period and $\text{sgn}(\cdot)$ is the sign of its argument. In the period when only TMNS is present, the synthetic multi-modal variable should ideally show negative response, since all the physio-markers are expected to decrease and all the physio-marker weights α_i , $i = 1, \dots, N_y$ are greater than or equal to zero. Hence, we defined J_3 as follows:

$$J_3 = \frac{1}{N_3} \sum_{k=1}^{N_3} \text{sgn}(\phi(y(k))) \phi^2(y(k)) \quad (6)$$

where N_3 is the number of data points in the TMNS period.

To solve the optimization problem in Eq. (2), the weights w_1 , w_2 , and w_3 must be specified. To select the optimal tuple of these weights, we repetitively solved the optimization problem in Eq. (2) pertaining to a large number of candidate weights (while their sum was constrained to unity: $w_1 + w_2 + w_3 = 1$) in a leave-one-subject-out (LOSO) setting. For a given tuple of weights, we used the physio-marker data from 39 experiments (except the held-out experiment) to solve Eq. (2). We set the initial condition for the physio-marker weights (i.e., α_i , $i = 1, \dots, N_y$ in Eq. (1)-(2)) as the 1st principal component of the six physio-markers pertaining to the 39 experiments (derived from principal components analysis; according to our prior work, the 1st principal component is a reasonable option to linearly combine multiple physio-markers

to capture the sympathetic arousal to acute mental stress [39]). Then, we calculated the time series sequence of the synthetic multi-modal variable ϕ pertaining to the held-out experiment by multiplying the optimal physio-marker weights (i.e., α_i , $i = 1, \dots, N_y$ in Eq. (1)-(2)) derived from the optimization to the time series sequences of the held-out physio-markers (i.e., $y = \{\text{HR}, \text{PTT}^{-1}, \text{PAT}^{-1}, \text{PEP}^{-1}, \text{A}_{\text{PPG}}^{-1}, \text{HR} \cdot \text{LVET}\}$). Then, we calculated the following metrics:

1) Explainability: An experiment is classified as explainable if the average value of its synthetic multi-modal variable is (i) positive in the acute mental stress-TMNS period and (ii) negative in the TMNS period.

2) Consistency: Consistency pertaining to a particular period is defined as the percentage of explainable synthetic multi-modal variable values in the period (i.e., positive values in the acute mental stress-TMNS period and negative values in the TMNS period).

3) Sensitivity: Sensitivity pertaining to a particular period is defined as the average value of synthetic multi-modal variable in the period. Considering that synthetic multi-modal variable response in the TMNS period is expected to be negative, we defined the sensitivity in this period as the negated average value of synthetic multi-modal variable in the period.

Note that we excluded the initial 30s in both the periods to remove the effect of moving-average filtering (conducted as part of post-processing; see Section II.A and Fig. 1). After solving Eq. (2) 40 times for all the experiments, we selected the tuple of weights that harmoniously maximized all the above metrics, i.e., the number of explainable experiments, average consistency in the acute mental stress-TMNS period and in the TMNS period, and average sensitivity in the acute mental stress-TMNS period and in the TMNS period. Given that these metrics can be competing against each other, we visually inspecting how the metrics depend on the weights and then selected the tuple of weights. Then, we solved Eq. (2) with Eq. (3) as the cost function with the weights thus selected to derive the optimal physio-marker weights for all the 40 experiments in the LOSO setting described above. Then, we calculated the time series sequence of the synthetic multi-modal variable pertaining to each experiment by multiplying the optimal physio-marker weights derived from the remaining 39 experiments to the time series sequences of the physio-markers pertaining to the experiment. We used MATLAB and its Optimization Toolbox to carry out all the computations.

C. Comparison of Synthetic Multi-Modal Variable vs Individual Physio-Markers

To determine the strengths and weaknesses of the synthetic multi-modal variable in comparison with individual physio-markers, we compared them in terms of a number of quantitative metrics applied equally to both the synthetic multi-modal variable and the individual physio-markers in isolation: (i) the number of explainable experiments, (ii) consistency, (iii) balanced sensitivity, defined as the ratio between the sensitivity in the acute mental stress-TMNS period and the sensitivity in the TMNS period, and (iv) robustness, defined as the coefficient of variation (CoV)

TABLE I. Number of explainable experiments pertaining to the individual physio-markers in isolation and the population-based synthetic multi-modal variable.

HR	PEP ⁻¹	A _{PPG} ⁻¹	PAT ⁻¹	PTT ⁻¹	HR-LVET	ϕ
11	9	19	10	17	14	24

TABLE II. Consistency, balanced sensitivity, and robustness metrics pertaining to all the individual physio-markers and the population-based synthetic multi-modal variable, both associated with all the 40 experiments (median (IQR)). *: significantly different from ϕ ($p < 0.05$ with Holm-Bonferroni correction, Wilcoxon signed rank test). AMS: acute mental stress.

	Consistency [%] (AMS-TMNS)	Consistency [%] (TMNS)	Balanced Sensitivity	Robustness (AMS-TMNS)	Robustness (TMNS)
HR	39 (1-82)*	66 (43-87)	-0.18 (-2.09-2.37)	-0.27 (-0.55-0.67)	0.65 (-1.50-1.30)
PEP ⁻¹	100 (59-100)	34 (1-64)*	-1.22 (-3.01-0.94)	0.25 (0.16-0.44)	-0.46 (-1.46-0.49)
A _{PPG} ⁻¹	100 (54-100)	81 (41-100)	0.29 (-3.10-1.92)	0.22 (0.09-0.47)	0.25 (-0.87-0.80)
PAT ⁻¹	100 (88-100)	50 (11-80)*	-1.06 (-4.75-0.98)	0.19 (0.08-0.28)*	-0.37 (-1.60-0.93)
PTT ⁻¹	100 (48-100)	59 (24-89)	-0.14 (-2.40-2.02)	0.30 (0.08-0.58)	0.43 (-0.71-1.72)
HR-LVET	88 (37-100)	58 (35-79)*	-0.39 (-1.94-1.02)	0.35 (-0.39-0.76)	0.49 (-2.41-1.23)
ϕ	100 (58-100)	83 (49-100)	0.73 (-0.95-2.83)	0.28 (0.14-0.56)	0.48 (-0.09-1.08)

TABLE III. Consistency, balanced sensitivity, and robustness metrics pertaining to all the individual physio-markers and the population-based synthetic multi-modal variable, both associated with the explainable experiments (median (IQR)). *: significantly different from ϕ ($p < 0.05$ with Holm-Bonferroni correction, Wilcoxon rank sum test). AMS: acute mental stress.

	Consistency [%] (AMS-TMNS)	Consistency [%] (TMNS)	Balanced Sensitivity	Robustness (AMS-TMNS)	Robustness (TMNS)
HR	84 (69-99)	86 (61-94)	2.48 (0.41-6.17)	0.84 (0.63-1.63)	0.93 (0.63-6.90)
PEP ⁻¹	100 (98-100)	77 (64-88)	2.16 (1.60-4.50)	0.29 (0.20-0.80)	1.95 (0.87-2.68)
A _{PPG} ⁻¹	100 (87-100)	87 (82-100)	1.90 (1.08-5.02)	0.41 (0.19-0.81)	0.74 (0.27-1.28)
PAT ⁻¹	100 (100-100)	82 (74-88)	2.95 (1.25-4.90)	0.29 (0.16-0.43)	1.09 (0.90-2.12)
PTT ⁻¹	100 (75-100)	78 (65-93)	2.11 (0.54-4.38)	0.39 (0.23-1.41)	1.02 (0.71-2.05)
HR-LVET	97 (88-100)	77 (63-92)	1.57 (0.52-5.92)	0.62 (0.29-0.88)	1.03 (0.63-2.48)
ϕ	100 (92-100)	85 (77-99)	2.14 (1.01-3.47)	0.42 (0.21-0.84)	0.77 (0.50-1.51)

pertaining to the time series sequence of the response of interest (either synthetic multi-modal variable or individual physio-markers), in the acute mental stress-TMNS period and in the TMNS period (in this way, smaller CoV means higher robustness). Note that we excluded the initial 30s in both the periods when calculating these quantitative metrics to remove the effect of moving-average filtering (conducted as part of post-processing; see Section II.A and Fig. 1).

D. Comparison of Population-Based vs Personalized Synthetic Multi-Modal Variables

To determine the potential advantage of personalizing the synthetic multi-modal variable to each subject, we compared population-based vs personalized synthetic multi-modal variables. For this purpose, we developed synthetic multi-modal variables specific to 40 individual experiments by solving the optimization problem in Section II.B with the data pertaining only to the individual experiments. To solve the optimization problem, we selected the tuple of weights in the cost function in Eq. (3) that harmoniously maximized consistency and sensitivity pertaining to the combined acute mental stress and TMNS responses as well as the TMNS response alone. Then, we compared the population-based vs personalized synthetic multi-modal variables in terms of the metrics defined in Sections II.B-II.C, namely: explainability, consistency, balanced sensitivity, and robustness, all pertaining to (i) all the 40 experiments and (ii) explainable experiments.

III. RESULTS

Fig. 3 shows the responses of the six physio-markers and the population-based synthetic multi-modal variable to (i) simultaneous acute mental stress and TMNS and (ii) TMNS

alone, pertaining to (a) all the experiments and (b) those experiments in which acute mental stress and TMNS responses were both explainable. Table I summarizes the number of explainable experiments pertaining to the individual physio-markers and the population-based synthetic multi-modal variable. Table II summarizes the consistency, balanced sensitivity, and robustness metrics pertaining to all the individual physio-markers and the population-based synthetic multi-modal variable, both associated with all the 40 experiments. Table III summarizes the same metrics pertaining to all the individual physio-markers and the population-based synthetic multi-modal variable, both associated with the explainable experiments. Fig. 4-6 respectively compare the consistency, balanced sensitivity, and robustness metrics pertaining to the six physio-markers and the population-based synthetic multi-modal variable, pertaining to (a) all the experiments and (b) those experiments in which acute mental stress and TMNS responses were both explainable. Fig. 7 shows the responses of the personalized synthetic multi-modal variable to (i) simultaneous acute mental stress and TMNS and (ii) TMNS alone, pertaining to (a) all the experiments and (b) those experiments in which acute mental stress and TMNS responses were both explainable. Fig. 8-10 respectively compare the consistency, balanced sensitivity, and robustness metrics pertaining to population-based vs personalized synthetic multi-modal variables, pertaining to (a) all the experiments and (b) those experiments in which acute mental stress and TMNS responses were both explainable.

IV. DISCUSSION

Effective management of acute mental stress calls for the means to both monitor physiological responses to such stress

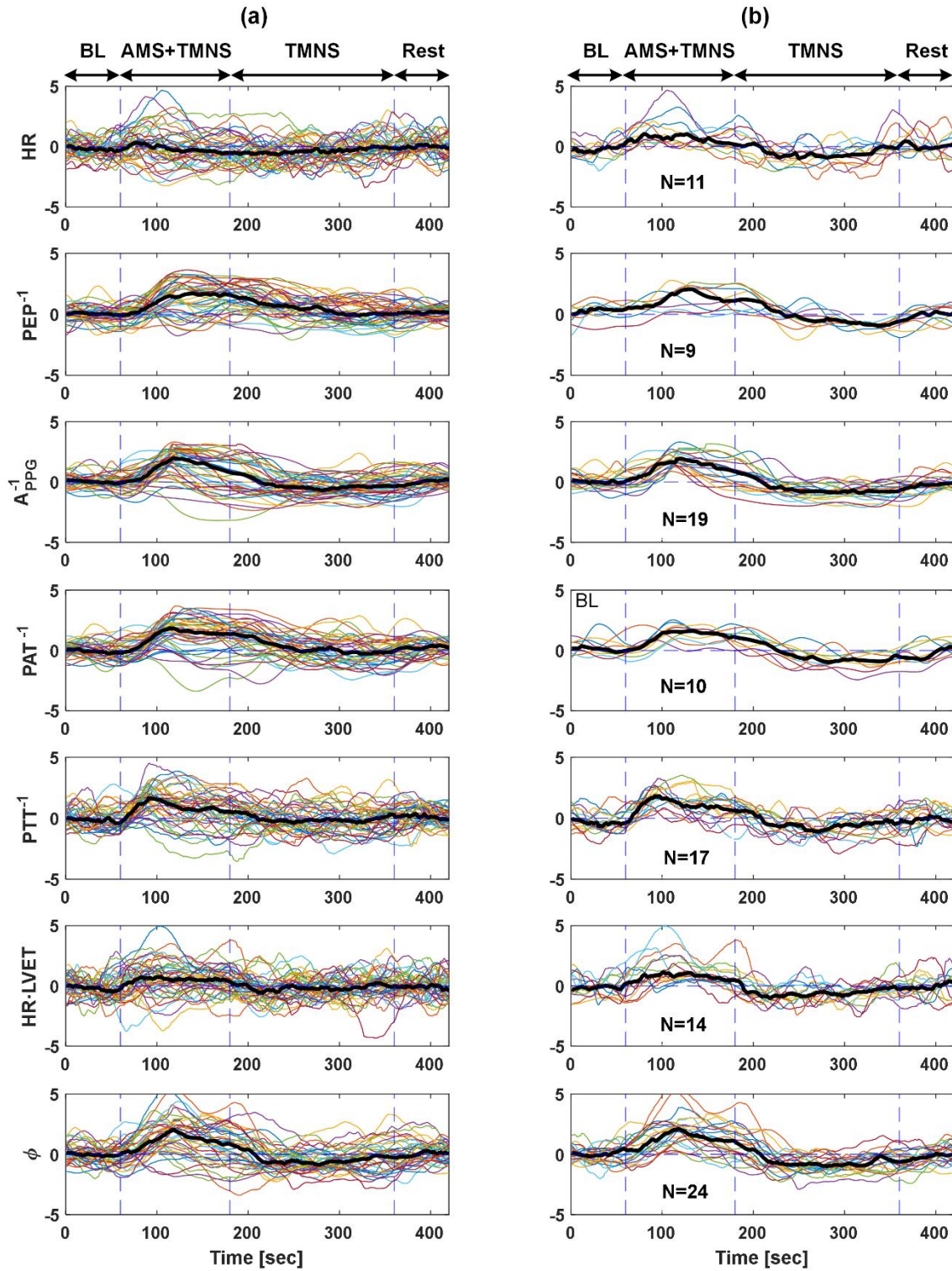


Fig. 3. Responses of the six physio-markers and the population-based synthetic multi-modal variable to (i) simultaneous acute mental stress and TMNS and (ii) TMNS alone, pertaining to (a) all the experiments and (b) those experiments in which acute mental stress and TMNS responses were both explainable.

and peripheral neuromodulation. Now that acute mental stress often elicits cardiovascular responses, it may be monitored by physio-markers representing a wide range of cardiovascular responses. In addition, transcutaneous neuromodulation, e.g., TMNS, has the potential to mitigate sympathetic arousal to acute mental stress. Hence, continuous physio-marker monitoring combined with TMNS may enable closed-loop

mitigation of acute mental stress. However, the best method

to capture the stress-mitigating effect of TMNS is unknown.

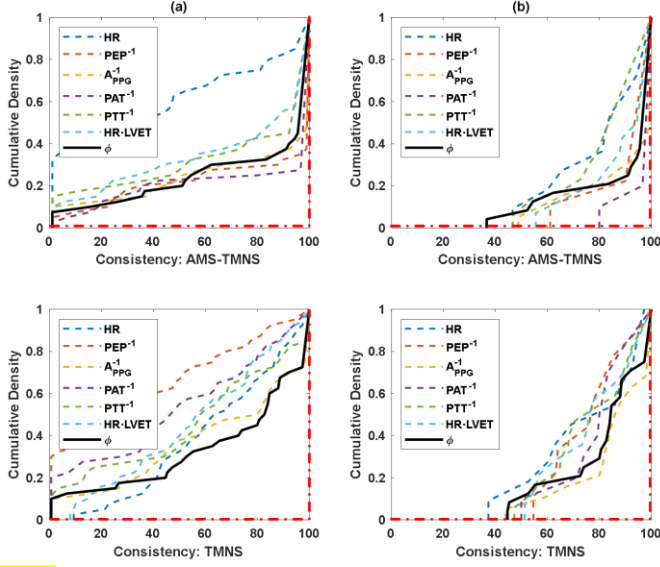


Fig. 4. Consistency metric pertaining to the six physio-markers and the population-based synthetic multi-modal variable, pertaining to (a) all the experiments and (b) those experiments in which acute mental stress and TMNS responses were both explainable. Ideal cumulative density is shown as red dash dot line.

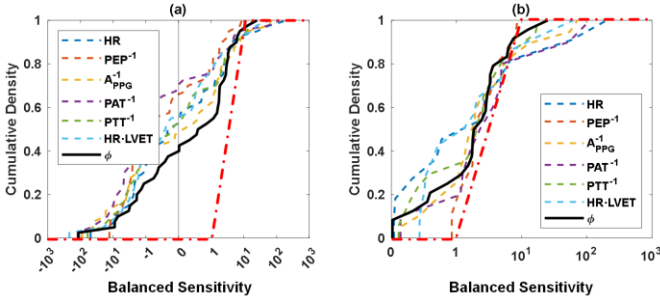


Fig. 5. Balanced sensitivity metric pertaining to the six physio-markers and the population-based synthetic multi-modal variable, pertaining to (a) all the experiments and (b) those experiments in which acute mental stress and TMNS responses were both explainable. Ideal cumulative density is shown as red dash dot line.

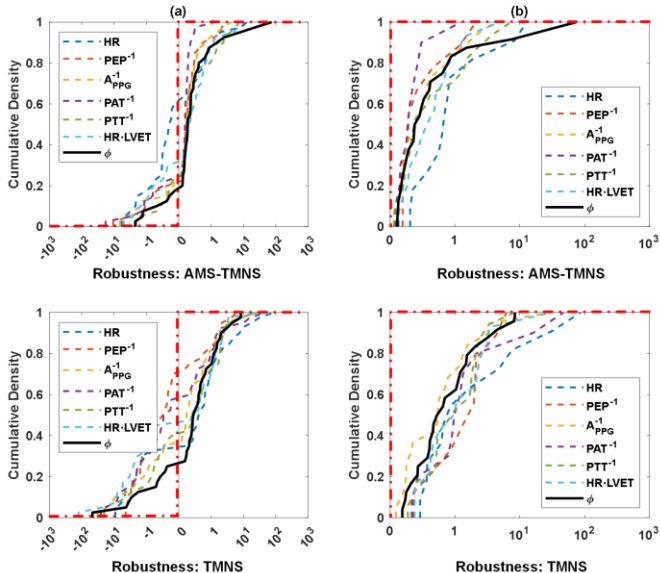


Fig. 6. Robustness metric pertaining to the six physio-markers and the population-based synthetic multi-modal variable, pertaining to (a) all the experiments and (b) those experiments in which acute mental stress and TMNS responses were both explainable. Ideal cumulative density is shown as red dash dot line.

Motivated by the hypothesis that the stress-mitigating effect of TMNS may be captured by the physio-markers which respond sensitively to acute mental stress, we developed and evaluated a synthetic multi-modal variable to measure cardiovascular responses to acute mental stress and TMNS using six plausibly explainable physio-markers we presented in our prior work [14]. Although the existing body of literature demonstrates that sympathetic arousal to acute mental stress can be captured based on various physiological signals, most prior work has used a restricted number of simple physio-markers derived from individual physiological signals without cross-integration (e.g., HR and its variability derived from the ECG or the PPG [1], [15], [40], [41], [42], [43], [44] as well as skin conductance response derived from the electrodermal activity signal [1], [15], [40], [41]). Notably, more recent work has started to explore cross-integration of multi-modal physiological signals (e.g., PAT derived from the ECG and the PPG [45], [46]). In this context, our idea to fuse multiple plausibly explainable physio-markers (including those leveraging cross-integration such as PEP using the ECG and the SCG, PTT using the SCG and the PPG, and PAT using the ECG and PPG) to develop a novel synthetic multi-modal variable to capture cardiovascular responses to acute mental stress and TMNS is innovative as well as timely. In particular, our work demonstrated, perhaps for the first time to the best of our knowledge, that multi-modal physiological signals have the potential to explainably quantify cardiovascular responses to TMNS.

A. Synthetic Multi-Modal Variable vs Individual Physio-Markers: Performance

Individual physio-markers showed limitations in capturing cardiovascular responses to both acute mental stress and TMNS (Table I and Fig. 4-6). In contrast, the synthetic multi-modal variable appeared superior to individual physio-markers in many aspects as follows:

1) Explainability: An ideal variable capturing cardiovascular responses to acute mental stress and TMNS must exhibit explainable behaviors in as many experiments as possible. The synthetic multi-modal variable showed explainable responses in 24 experiments (thus 60%), which was higher than all the individual physio-markers which showed explainable responses only in ≤ 19 experiments (thus, $< 48\%$) (Table I). Hence, the synthetic multi-modal variable outperformed all the individual physio-markers in terms of explainability.

2) Consistency: An ideal variable capturing cardiovascular responses to acute mental stress and TMNS must maintain explainability through time, i.e., with 100% consistency. This translates into an ideal cumulative density pertaining to consistency vs experiment, which is zero throughout except at 100% when it attains 1 (i.e., all the experiments must have 100% consistency). Considering all the experiments (Fig. 4(a)), the synthetic multi-modal variable outperformed (i) HR, PTT⁻¹, and HR-LVET in the acute mental stress-TMNS period and (ii) all the individual physio-markers in the TMNS period. Considering only explainable experiments (Fig. 4(b)), the

synthetic multi-modal variable outperformed (i) most individual physio-markers except PAT^{-1} and A_{PPG}^{-1} in the acute mental stress-TMNS period and (ii) all the individual

small number of extremely large balanced sensitivity values relative to most individual physio-markers. Notably, HR, PAT^{-1} , A_{PPG}^{-1} , and $HR \cdot LVET$ suffered from a large number of

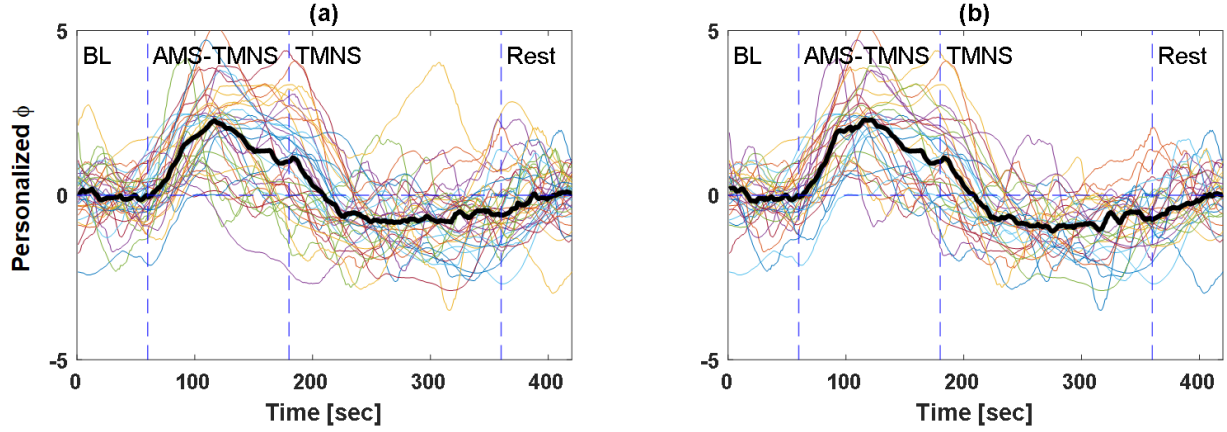


Fig. 7. Responses of the personalized synthetic multi-modal variable to (i) simultaneous acute mental stress and TMNS and (ii) TMNS alone, pertaining to (a) all the experiments (i.e., 40) and (b) those experiments in which acute mental stress and TMNS responses were both explainable (i.e., 34).

TABLE IV. Consistency, balanced sensitivity, and robustness metrics pertaining to population-based and personalized synthetic multi-modal variables associated with all the 40 experiments (median (IQR)). *: significantly different from population-based ϕ ($p < 0.05$, Wilcoxon signed rank test). AMS: acute mental stress.

	Consistency [%] (AMS-TMNS)	Consistency [%] (TMNS)	Balanced Sensitivity	Robustness (AMS-TMNS)	Robustness (TMNS)
Population ϕ	100 (58-100)	83 (49-100)	0.73 (-0.95-2.83)	0.28 (0.14-0.56)	0.48 (-0.09-1.08)
Personalized ϕ	100 (83-100)	83 (64-98)	1.56 (0.34-3.82)*	0.28 (0.17-0.86)	0.57 (0.47-1.85)*

TABLE V. Consistency, balanced sensitivity, and robustness metrics pertaining to population-based and personalized synthetic multi-modal variables associated with the explainable experiments (median (IQR)). *: significantly different from population-based ϕ ($p < 0.05$, Wilcoxon rank sum test). AMS: acute mental stress.

	Consistency [%] (AMS-TMNS)	Consistency [%] (TMNS)	Balanced Sensitivity	Robustness (AMS-TMNS)	Robustness (TMNS)
Population ϕ	100 (92-100)	85 (77-99)	2.14 (1.01-3.47)	0.42 (0.21-0.84)	0.77 (-0.50-1.51)
Personalized ϕ	100 (83-100)	91 (68-99)	2.00 (0.78-3.99)	0.36 (0.19-0.97)	0.74 (0.50-2.35)

physio-markers except A_{PPG}^{-1} in the TMNS period.

3) **Balanced sensitivity:** An ideal variable capturing cardiovascular responses to acute mental stress and TMNS must be responsive to both of them. Our experimental data clearly suggest that cardiovascular responses to acute mental stress tend to be larger in magnitude than those to TMNS (Fig. 3). Hence, balanced sensitivity of an ideal variable must be associated with an ideal cumulative density which is zero at negative values (i.e., explainable) and steeply increases in a range close to unity (e.g., 1-10, meaning that the magnitude of TMNS response is within 0.1-1.0 times the magnitude of acute mental stress-TMNS response). Considering all the experiments (Fig. 5(a)), the synthetic multi-modal variable was associated with the smallest negative balanced sensitivity values (meaning it is the most explainable), and it also exhibited a steep increase in the range of 1-10. In contrast, all the individual physio-markers suffered from a large proportion of negative balanced sensitivity values (approximately 40%-70%). Notably, PAT^{-1} had the largest density pertaining to negative values, which weakens its advantage in consistency. Considering only explainable experiments (Fig. 5(b)), the synthetic multi-modal variable exhibited the steepest increase in its cumulative density relative to all the individual physio-markers within the range of 1-10. In addition, it involved a

extremely large balanced sensitivity values, which weakens the advantage of PAT^{-1} and A_{PPG}^{-1} in consistency in particular.

4) **Robustness:** An ideal variable capturing cardiovascular responses to acute mental stress and TMNS must exhibit small fluctuations in the steady state, i.e., small CoV. This translates into an ideal cumulative density pertaining to robustness vs experiment, which is zero at negative values (i.e., explainable) and one at positive values with a jump at $CoV=0$. Considering all the experiments (Fig. 6(a)), the synthetic multi-modal variable outperformed most if not all the individual physio-markers in terms of the proportion of unexplainable experiments in both acute mental stress-TMNS and TMNS periods. In addition, the synthetic multi-modal variable attained the cumulative density of 1 comparably to all the individual physio-markers in the TMNS period while it showed a modestly sluggish behavior in the acute mental stress-TMNS period (due to a small number of experiments (=3) in which the synthetic multi-modal variable exhibited large sustained swings in its response; see Fig. 3). Considering only explainable experiments (Fig. 6(b)), the synthetic multi-modal variable outperformed all the individual physio-markers except A_{PPG}^{-1} in the TMNS period, while it showed a modestly sluggish behavior in the acute mental stress-TMNS period due to the three above-mentioned

experiments with large sustained swings.

Taking all the above metrics into account, the synthetic

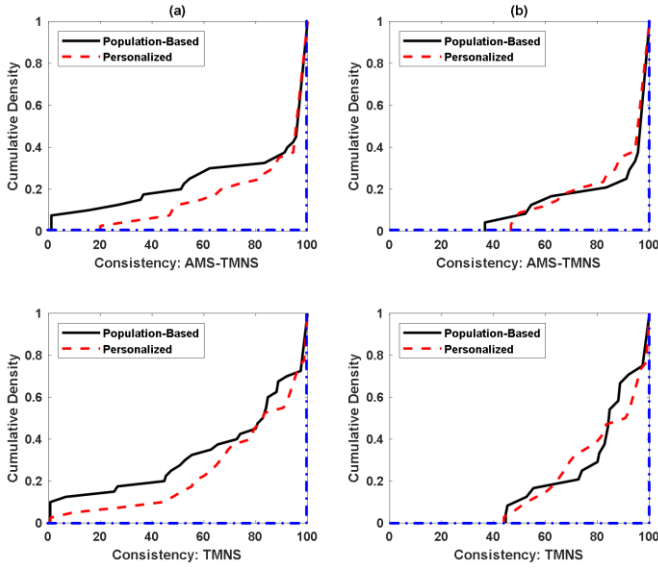


Fig. 8. Consistency metric pertaining to population-based vs personalized synthetic multi-modal variables, pertaining to (a) all the experiments (i.e., 40) and (b) those experiments in which acute mental stress and TMNS responses were both explainable (24 population-based vs 34 for personalized). Ideal cumulative density is shown as blue dash dot line.

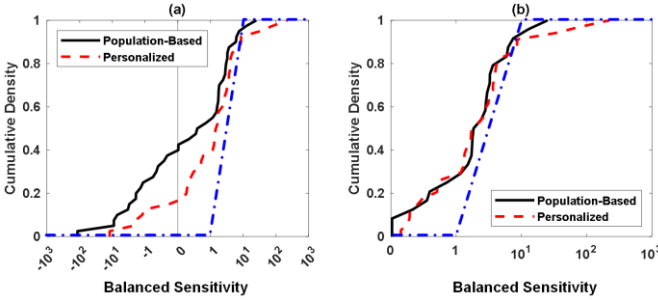


Fig. 9. Balanced sensitivity metric pertaining to population-based vs personalized synthetic multi-modal variables, pertaining to (a) all the experiments (i.e., 40) and (b) those experiments in which acute mental stress and TMNS responses were both explainable (24 population-based vs 34 for personalized). Ideal cumulative density is shown as blue dash dot line.

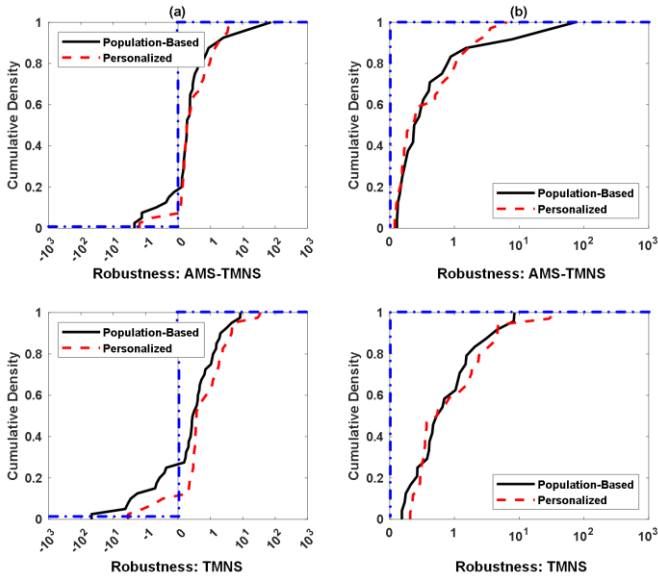


Fig. 10. Robustness metric pertaining to population-based vs personalized synthetic multi-modal variables, pertaining to (a) all the experiments (i.e., 40) and (b) those experiments in which acute mental stress and TMNS responses were both explainable (24 population-based vs 34 for personalized). Ideal cumulative density is shown as blue dash dot line.

multi-modal variable was superior to all the individual physio-markers all in all. It was superior to all the individual physio-markers in terms of explainability. It was superior to (i) HR, PTT⁻¹, and HR·LVET in terms of consistency, balanced sensitivity, and robustness; (ii) PAT⁻¹ and PEP⁻¹ in terms of balanced sensitivity and robustness; and (iii) A_{PPG}⁻¹ in terms of balanced sensitivity with comparable consistency and robustness. However, most of the differences between the synthetic multi-modal variable and the individual physio-markers were not statistically significant (Table II and Table III). This may be attributed to a few possible reasons, e.g., (i) small sample size and (ii) the limited efficacy of population-based synthetic multi-modal variable due to a large inter-individual variability in the physio-marker responses to acute mental stress and TMNS. Nonetheless, the results suggest that judicious fusion of multi-modal physio-markers may have advantages in capturing the cardiovascular responses to acute mental stress and TMNS more effectively than individual physio-markers.

B. Synthetic Multi-Modal Variable: Physio-Marker Composition

The synthetic multi-modal variable was composed primarily of A_{PPG}⁻¹ (81%), HR (11%), and HR·LVET (6%). The dominance of A_{PPG}⁻¹ is not surprising, noting that it is the most powerful physio-marker among the six considered in this work in terms of the metrics used in this work: explainability, consistency, balanced sensitivity, and robustness (Fig. 3-Fig. 6). Notably, A_{PPG}⁻¹ responded well to transcutaneous cervical vagus nerve stimulation in our prior work [47]. In contrast, PAT⁻¹ and PEP⁻¹ contributed minimally to the synthetic multi-modal variable. These physio-markers exhibited strong responses to acute mental stress but highly variable (including unexplainable) responses to TMNS (Fig. 3). Remarkably, HR and HR·LVET in our experimental data were strongly correlated with $r=0.78$ on the average. Hence, the synthetic multi-modal variable may practically be calculated using only A_{PPG}⁻¹ and HR. Given that both PPG amplitude and HR can be extracted from the PPG, the synthetic multi-modal variable may be easily implemented using a PPG signal alone, e.g., using a wristwatch equipped with a PPG sensor. In this way, the synthetic multi-modal variable has the potential to enable unobtrusive and ubiquitous monitoring of cardiovascular responses to acute mental stress and TMNS.

C. Population-Based vs Personalized Synthetic Multi-Modal Variable

The composition of the synthetic multi-modal variable exhibited large variability when personalized to individual experiments, reflecting diverse physio-marker responses to acute mental stress and TMNS (not shown). Upon personalization, the synthetic multi-modal variable exhibited explainable responses in 34 experiments (thus 85%) (Fig. 7). In terms of all the metrics (consistency, balanced sensitivity, and robustness), the personalized synthetic multi-modal variable outperformed its population-based counterpart in both the acute mental stress-TMNS period and the TMNS period when all the experiments were considered (Fig. 8(a), Fig. 9(a), and Fig. 10(a)). In addition, the former was comparable to the

latter in both the acute mental stress-TMNS period and the TMNS period when only the explainable experiments were considered (Fig. 8(b), Fig. 9(b), and Fig. 10(b)), which is notable because the former includes more explainable experiments than the latter (34 vs 24). Table IV and Table V summarize these metrics pertaining to population-based vs personalized synthetic multi-modal variables, both associated with all the 40 experiments and the explainable experiments, respectively. All in all, the results suggest that cardiovascular responses to acute mental stress and TMNS are associated with a large inter-individual variability, and that a synthetic multi-modal variable based on a population-based physio-marker composition may not achieve a universally effective capture of these diverse responses. Hence, personalizing the physio-marker composition may have additional value on top of fusing physio-markers in monitoring acute mental stress-TMNS responses. However, personalization of the synthetic multi-modal variable will also necessitate novel algorithms and protocols to enable online learning and adaptation of the synthetic multi-modal variable (its composition weights α_i , $i = 1, \dots, N_y$ in particular).

D. Limitations

This work has several limitations to be addressed. First, the number of participants is small ($N=20$). Hence, the physio-marker composition in the synthetic multi-modal variable may change if the experimental data are expanded to include more experiments and participants. However, the method presented in this work may remain valid and applicable. Second, it remains unclear if the six physio-markers used in this work are optimally effective in capturing TMNS responses. Regardless, given that (i) they are effective in capturing acute mental stress responses and that (ii) their responses must normalize upon the blunting of acute mental stress, the physio-markers may still be reasonable if not optimal. However, follow-up efforts to discover ideal physio-markers representing TMNS responses may be rewarding. Third, the experimental protocol may not have been ideally suited to this work. Especially, the experimental protocol included periods containing acute mental stress and TMNS as well as TMNS alone but not a period containing acute mental stress alone. Having an acute mental stress period followed by an acute mental stress-TMNS period may have made it easier to highlight the ability of TMNS to mitigate acute mental stress responses. The protocol employed in our experimental data intended to reveal TMNS responses in the presence vs absence of acute mental stress, which is meaningful in itself. But, it is possible that TMNS responses may be clearer in the presence of acute mental stress than in its absence (since in the absence of acute mental stress, the stress-mitigating effect of TMNS may be marginal). Further, the experimental data were collected in a cross-over study to compare multiple transcutaneous peripheral nerve stimulation modalities, and TMNS was not always the first stimulation that the participants received. Hence, the responses to acute mental stress and TMNS may be associated with habituation effect. Hence, follow-up efforts to explicitly investigate the stress-mitigating effect of TMNS may be rewarding.

V. CONCLUSION

We demonstrated that judicious fusion of multi-modal physio-markers of sympathetic arousal to acute mental stress may enable the development of a synthetic multi-modal variable capable of capturing cardiovascular responses to acute mental stress and TMNS superior to individual physio-markers. Future work must confirm the findings in this work using a large dataset with an enhanced experimental protocol. In addition, investigations to leverage the synthetic multi-modal variable in closed-loop controlled mitigation of acute mental stress are necessary.

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